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Systems for treating an installed and assembled gas turbine engine

Abstract

Systems and methods for treating a component of an installed and assembled gas turbine engine are provided. Accordingly, the system includes a storage vessel configured for containing a treating fluid. The system further includes a delivery assembly operably coupled to the storage vessel and a control unit. The delivery assembly is configured to atomize a portion of the treating fluid to develop a treating mist of atomized droplets having a median diameter facilitating suspension of the atomized droplets within at least one flow path of the core gas turbine engine. A flow generation assembly of the system is configured to facilitate a passage of the treating mist along the at least one flow path of the core gas turbine engine so as to wet and treat the component.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. application Ser. No. 17/220,112, filed on Apr. 1, 2021, now U.S. Pat. No. 11,555,413, which is a continuation-in-part of U.S. application Ser. No. 17/027,754, filed on Sep. 22, 2020, now U.S. Pat. No. 11,371,425, both of which are hereby incorporated by reference herein in their entirety.

FIELD

(1) The present subject matter relates generally to gas turbine engines, and more particularly, to systems and methods for treating components of assembled gas turbine engines on-wing.

BACKGROUND

(2) A typical aircraft-mounted gas turbine engine includes gas turbine engine components having very fine cooling passages that allow for higher gas temperatures in the combustor and/or the high-pressure or low-pressure turbines. During operation, particularly in environments that contain fine-scale dust, environmental particulate accumulates on engine components and within the cooling passages of the engine. For example, dust (reacted or non-reacted), sand, or similar, can build up on the flow path components and on the component, internal, impingement cooled surfaces during turbine engine operation. In addition, particulate matter entrained in the air that enters the turbine engine and the cooling passages can contain sulfur-containing species that can corrode the components. Such deposits can lead to reduced cooling effectiveness of the components and/or corrosive reaction with the metals and/or coatings of the engine components. Thus, deposits can lead to premature distress, reduced engine life, and/or increased repair/maintenance costs (e.g. cost to restore components during an engine overhaul. Additionally, accumulations of environmental contaminants (e.g. dust-reacted and unreacted, sand, etc.) can degrade aerodynamic performance of the high-pressure components and lower fuel efficiency of the engine through changes in airfoil morphology and/or affect coatings applied to the engine components. As such, it may be desirable from time to time to treat the components of the assembled gas turbine engine on-wing. For example, treating the blades and vanes of the compressor may improve the efficiency of compression and result in a lower fuel consumption and/or lower exit temperature for the compressor. These, in turn, may result in a lower operating temperature of the hot section of the engine, which may increase the operating life of the various components.

(3) Accordingly, an improved system and method to treat gas turbine engine components would be useful.

BRIEF DESCRIPTION

(4) Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

(5) In one aspect, the present disclosure is directed to a method for treating at least one component of an installed and assembled gas turbine engine. The gas turbine engine may include a core gas turbine engine. The core gas turbine engine may have an absence of powered rotation during the treating. The method may include operably coupling a delivery assembly to an annular inlet of the core gas turbine engine. The delivery assembly may be coupled to a control unit and a storage vessel containing a treating fluid. The method may also include atomizing a portion of the treating fluid with the delivery assembly to develop a treating mist. The treating mist may include a plurality of atomized droplets. The method may include suspending the atomized droplets of the treating mist within an airflow within at least one flow path of the core gas turbine engine. The treating mist may occupy a cross-sectional area of the at least one flow path to establish a simultaneous cross-sectional contact therein. At least a portion of the atomized droplets may remain suspended within the flow path(s) from the annular inlet to an axial position downstream of a high-pressure compressor of the gas turbine engine. Additionally, the method may include impacting or precipitating a portion of the treating mist onto the component(s) so as to wet at least 80% of an exposed, inlet-facing surface of the component(s).

(6) In an embodiment, the atomized droplets may have a median diameter of less than or equal to 5 microns and the treating mist has a fluid-to-air mass ratio of at least 0.025 and less than or equal to 5.0 kilograms of treating fluid to kilograms of air.

(7) In an additional embodiment, a thermal state of the core gas turbine engine may be less than or equal to 135% of an ambient air absolute temperature and the axial position downstream of the high-pressure compressor may be an axial position downstream of the installed and assembled gas turbine engine.

(8) In a further embodiment, the method may include establishing an elevated delivery temperature of the treating mist which increases the vapor content of the treating mist within the flow path(s).

(9) In an embodiment, the method may include supplying a surge portion of the treating mist to the core gas turbine engine. The method may also include operably decoupling the delivery assembly from the annular inlet. Further, the method may include establishing a soak period during which the treating fluid affects at least one of the component(s) and a deposit thereon.

(10) In an additional embodiment, the delivery assembly may include an array of nozzles. Each nozzle of the array of nozzles may be configured to develop atomized droplets having a median diameter of less than or equal to 5 microns. The method may also include actuating at least one nozzle of the array of nozzles to establish a treating mist volume. The treating mist volume may include a concentration of atomized droplets within a specified portion of the flow path(s). The treating mist volume may be characterized by a fluid-to-air mass ratio of at least 0.025 and less than or equal to 5.0 kilograms of treating fluid to kilograms of air.

(11) In a further embodiment, the method may include obtaining environmental data. The environmental data may include an ambient temperature, an ambient pressure, and an ambient humidity affecting the installed and assembled gas turbine engine. The method may also include obtaining data indicative of a thermal state of the core gas turbine engine. Based on the environmental data and the data indicative of the thermal state of the core gas turbine engine, the method may include establishing the treating mist volume delivered to the annular inlet. Establishing the treating mist volume may also include establishing a treating mist flow rate within the flow path(s).

(12) In an embodiment, the method may include monitoring an absolute humidity level at a point-of-departure from the installed and assembled gas turbine engine. The method may also include utilizing the monitored humidity level to determine a percentage of the delivered treating mist remaining suspended at the point-of-departure from the installed and assembled gas turbine engine. Additionally, the method may include adjusting the treating mist volume delivered to the annular inlet based on the determined percentage so as to achieve a desired level of wetting of the component(s).

(13) In an additional embodiment, the method may include determining a first volume of treating fluid atomized and delivered as the treating mist to the annular inlet. The method may also include determining a second volume of treating fluid suspended at the point-of-departure from the installed and assembled gas turbine engine based on the monitored humidity level. Additionally, the method may include computing the portion of the first volume of treating fluid precipitated onto the component(s) based on a difference between the first and second volumes. Further, the method may include adjusting the treating mist volume delivered to the annular inlet so that the second volume is less than or equal to 35% of the first volume.

(14) In a further embodiment, the delivery assembly may include an array of nozzles. Additionally, operably coupling the delivery assembly to the annular inlet may also include positioning the array of nozzles within a fan section of the gas turbine engine so that an outlet of each nozzle of the array of nozzles is arranged at an axial location between the fan section and the annular inlet of the core gas turbine engine. Additionally, the method may include circumscribing the array of nozzles with at least one blocking element. The blocking element(s) may at least partially occlude an alternative flow path.

(15) In an embodiment, wherein treating the component(s) of the installed and assembled gas turbine engine is repeated at least once every 30 days.

(16) In an additional embodiment, the method may include establishing a treating cycle. The treating cycle may have a duration of 30 minutes or less. The method may also include operably decoupling the delivery assembly from the annular inlet by a conclusion of the treating cycle.

(17) In a further embodiment, atomizing a portion of the treating fluid may include atomizing less than 120 liters of treating fluid.

(18) In yet a further embodiment, the treating fluid may be a cleaning fluid, and the treating mist may be a cleaning mist. The method may also include dissolving at least a portion of a deposit on

the component(s) so as to remove the deposit.

(19) In an embodiment, the treating fluid may be a corrosion inhibitor. The corrosion inhibitor may be a catalyst configured to react with at least one identified contaminant so as to inhibit corrosion of the component(s).

(20) In an additional embodiment, the treating fluid may be a sealant configured to preclude a contact between a contaminant and the component(s).

(21) In a further embodiment, atomizing the portion of the treating fluid may include atomizing the portion of the treating fluid via a vibrating mesh nebulizer so as to develop the plurality of atomized droplets having a median diameter of at least 2 microns and less than or equal to 5 microns.

(22) These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

(2) FIG. 1 shows a schematic cross-sectional view of one embodiment of a gas turbine engine and a system for treating a component of the gas turbine engine according to the present disclosure;

(3) FIG. 2 shows a schematic view of a portion of the gas turbine engine of FIG. 1 and an embodiment of the system for treating the component according to the present disclosure.

(4) FIG. 3 shows a schematic view of a portion of the gas turbine engine of FIG. 1 and an embodiment of the system for treating the component according to the present disclosure.

(5) FIG. 4 shows a schematic view of an embodiment of a system for treating the component according to the present disclosure; and

(6) FIG. 5 depicts an exemplary controller such as for use in the exemplary systems of FIGS. 1-4; and

(7) FIG. 6 illustrates a flow diagram of one embodiment of a method for treating a component of the gas turbine engine of FIG. 1 according to the present disclosure.

(8) Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION

(9) Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

(10) As used herein, the terms “first” and “second” may be used interchangeably to distinguish one component(s) from another and are not intended to signify location or importance of the individual component(s).

(11) As used herein, the term “vapor” refers to a substance in the gaseous state, as distinguished

from the liquid or solid state.

(12) The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows. As used herein, the fluid flow is in the direction encountered by the gas turbine engine in flight operations.

(13) As used herein, the phrases “constructed of CMCs” and “comprised of CMCs” shall mean component(s) substantially constructed of ceramic matrix composites (CMCs). More specifically, the CMC component(s) shall include more CMC material than just a layer or coating of CMC materials. For example, the component(s) constructed of CMCs may be comprised or constructed substantially or entirely of CMC materials, including greater than about 50, 60, 70, 80, 90, or 100 percent CMC material.

(14) Certain approaches for treating (e.g. cleaning, sealing, etc.) assembled engines may rely on foam or liquids. However, the physical characteristics of the treating medium may impose limits on the engine surfaces which may be affected. Additionally, certain approaches to treating engines on-wing may require the aircraft to be positioned in a maintenance area of an airfield and taken out of service for an undesirable length of time. For example, certain approaches may require that the aircraft be taken out of service for 4-9 hours and may require the disassembly of a portion of the engine, the presence of a crew in a cockpit, and/or an external power source to rotate the engine. As a result of these drawbacks, it may not be practical for these treating approaches to be practiced more than 2-4 times per year. The extended intervals between treatments, such as cleanings, may result in the engine being operated in a fouled condition for a greater percentage of the time and/or suffering damage due to corrosion.

(15) One or more of the above-identified issues may be addressed by the presently disclosed methods and systems. In particular, methods and systems are generally provided for treating (e.g. sealing and/or removing deposits from) components of an installed and assembled gas turbine engine. The methods of the present disclosure generally provide for introducing a treating mist into the annular inlet of the engine. The treating mist may include atomized particles of a treating fluid which are sized to permit at least a portion of the treating mist to remain suspended in a flow path of the core engine at least to a location downstream of a high-pressure compressor. As the treating mist flows along the flow path, the atomized particles may encounter various components of the engine. When encountering the various components, the atomized particles may impact the component and/or precipitate onto the component thereby wetting the component with the treating fluid. By utilizing the treating mist, the methods described herein may be accomplished in 30 minutes or less without requiring the disassembly or rotation of the engine. Additionally, the utilization of the treating mist to wet the components facilitates an efficient treating cycle utilizing less than 120 liters treating fluid.

(16) Referring now to the drawings, FIG. 1 illustrates a cross-sectional view of one embodiment of an installed and assembled gas turbine engine **100** configured as an installed and assembled gas turbine engine. The engine **100** may be utilized with an aircraft in accordance with aspects of the present subject matter. However, the engine **100** may also be suitable for employment in electrical power generating applications, ship propulsion applications, and/or ground vehicle propulsion applications.

(17) The engine **100** is illustrated in FIG. 1 as having a longitudinal or axial centerline axis **112** extending therethrough for reference purposes. In an embodiment, the engine **100** may include at least one component **102** configured to affect a portion of ambient air entering the engine **100**. The component(s) **102** may include any of the elements of the engine **100** discussed herein. In an embodiment, the component(s) **102** may, for example, be formed from CMC, titanium, steel, aluminum, nickel, chromium and/or combinations thereof. Additionally, it should be appreciated that while the engine **100** is depicted in FIG. 1 as a turbofan jet engine, the systems and methods described herein may be employed on any turbomachine including, but not limited to, high-bypass

turbofan engines, low-bypass turbofan engines, turbojet engines, turboprop engines, turboshaft engines, propfan engines, and so forth.

(18) In general, the engine **100** may include a core gas turbine engine (indicated generally by reference character **114**) and a fan section **116** positioned upstream thereof. The core engine **114** may generally include a substantially tubular outer casing **118** that defines an annular inlet **120**. The annular inlet **120** may define an entrance to at least one flow path **104** of the core engine **114**.

(19) In an embodiment, the outer casing **118** may enclose and support a booster compressor **122** for increasing the pressure of the air that enters the core engine **114** to a first pressure level. A high-pressure (HP), multi-stage, axial-flow compressor **124** may be serially arranged at an axial position downstream of the booster compressor **122**. In operation, the HP compressor **124** may receive the pressurized air from the booster compressor **122** and further increase the pressure of such air.

(20) In an embodiment, a combustor **126** may be serially arranged at an axial position downstream of the HP compressor **124**. In operation, pressurized air exiting the HP compressor **124** may flow to the combustor **126** within which fuel may be injected by a fuel system **106** into the flow of pressurized air, with the resulting mixture being combusted within the combustor **126**.

(21) Referring still to FIG. **1**, in an embodiment, an HP turbine **128** may be serially arranged at an axial position downstream of the combustor **126** so that high energy combustion products may be directed from the combustor **126** along the flow path(s) **104** of the engine **100** to the HP turbine **128** for driving the HP compressor **124** via an HP driveshaft **130**.

(22) In an embodiment, a low-pressure (LP) turbine **132** may be serially arranged at an axial position downstream of the HP turbine **128**. The LP turbine **132** may be configured for driving the booster compressor **122** and the fan section **116** via an LP driveshaft **134**. In an embodiment, the LP driveshaft **134** may be generally coaxial with HP driveshaft **130**.

(23) As depicted in FIG. **1**, the flow path(s) **104** may communicatively couple the annular inlet **120** with a point-of-departure **136** from the engine **100**. For example, in operation, after driving the HP turbine **128** and the LP turbine **132**, the combustion products in the flow path(s) **104** may be expelled from the core engine **114** via the point-of-departure **136** configured as an exhaust nozzle to provide propulsive jet thrust.

(24) It should be appreciated that each turbine may generally include one or more turbine stages, with each stage including a turbine nozzle and a downstream turbine rotor. As will be described below, the turbine nozzle may include a plurality of vanes disposed in an annular array about the centerline axis **112** of the engine **100** for turning or otherwise directing the flow of combustion products through the turbine stage towards a corresponding annular array of rotor blades forming part of the turbine rotor. As is generally understood, the rotor blades may be coupled to a rotor disk of the turbine rotor, which is, in turn, rotationally coupled to the turbine's driveshafts **130**, **134**.

(25) Additionally, as shown in FIG. **1**, the fan section **116** of the engine **100** may generally include a rotatable, axial-flow fan rotor **138** surrounded by an annular fan casing **140**. In an embodiment, the LP driveshaft **134** may be operably coupled to the fan rotor **138**. It should be appreciated that the fan casing **140** may be supported relative to the core engine **114** by a plurality of substantially radially-extending, circumferentially-spaced outlet guide vanes **142**. As such, the fan casing **140** may enclose the fan rotor **138** and its corresponding fan rotor blades **144**. Moreover, a downstream section **146** of the fan casing **140** may extend over an outer portion of the core engine **114** to define at least one alternative flow path **148** between the fan casing **140** and the outer casing **118** of the core engine **114**. In an embodiment, the alternative flow path(s) **148** may provide additional propulsive jet thrust.

(26) As shown in FIGS. **1-4**, in an embodiment, a system **300** may be utilized to treat the component(s) **102** of the installed and assembled gas turbine engine **100** (e.g., an assembled, on-wing gas turbine engine). The system **300** may be employed when the core engine **114** is essentially stationary and not being rotated via the application of a power source. The system **300** may include a storage vessel **302** containing a treating fluid **304**. The system **300** may also include

a delivery assembly **306** operably coupled to the storage vessel **302** and a control unit **308**. The delivery assembly **306** may include at least one nozzle **310** configured to atomize a portion of the treating fluid **304** in order to develop a treating mist **312**. The treating mist **312** may include a plurality of atomized droplets. The atomized droplets may have a median diameter, and thus mass, which facilitates suspension of the atomized droplets within the flow path(s) **104** of the core engine **114**. In an embodiment, a portion of the atomized droplets of the treating mist **312** may be suspended within an airflow **318** within the flow path(s) **104** from the annular inlet **120** to an axial position (A) downstream of the HP compressor **124**. The treating mist **312** may occupy a cross-sectional area of the flow path(s) **104** to establish a simultaneous cross-sectional contact therein. In other words, in an embodiment, the treating mist **312** may essentially fill the flow path(s) **104** at a given axial location without requiring that the engine **100** be rotated. The system **300** may also include a flow generation assembly **314** which may be oriented to facilitate a passage of the treating mist **312** along the flow path(s) **104** of the core engine **114**. It should be appreciated that, in an embodiment, the engine **100** may be permitted to rotate at a rate not exceeding 5 rotations per minute (RPM) so as to facilitate specific treatment approaches (e.g., an extended cleaning and/or a treatment specifically directed to internal circuits of the cooling system). For example, in an embodiment, the engine **100** may be rotated at a rate of at least 0.05 RPM but less than or equal to 1.0 RPM.

(27) In an embodiment, the treating fluid **304** may include any suitable composition now known or later developed in the art. For example, in an embodiment, the treating fluid **304** may be configured as a cleaning fluid and may include a biodegradable citric and/or glycolic-acid composition including both ionic and non-ionic surfactants. Additionally, the treating fluid **304** may be compatible with all coatings and components **102** internal and external to the engine **100** and suitable for on-wing application. The treating fluid **304** may be utilized without requiring a rinse step prior to firing the engine **100** following the treating/cleaning. As a cleaning fluid, treating fluid **304** may demonstrate no pitting corrosion or intergranular attack to engine parent metals or coating systems. Accordingly, the treating fluid **304** may be a water-based cleaning fluid. For example, in an embodiment, the treating fluid **304** may be a water-detergent combination. In a further embodiment, the treating fluid **304** may be water without a detergent. Additionally, the water may be treated to remove potential contaminants, such as by distillation and/or deionization. It should be appreciated that when configured as a cleaning fluid, the treating fluid **304** may also include corrosion inhibition properties.

(28) In an additional embodiment, the treating fluid **304** may be configured as a corrosion inhibitor. In such an embodiment, the corrosion inhibitor may be a catalyst configured to react with at least one identified contaminant so as to inhibit corrosion of the component (s) **102**. For example, the treating fluid **304** may, in an embodiment, be a catalyst which reacts with sulfur. By reacting with sulfur, the treating fluid **304** may preclude an additional chemical reaction with the sulfur (or a compound thereof) which may corrode the component (s) **102**. Similarly, in an embodiment, the treating fluid **304** may be a catalyst which reacts with carbon so as to preclude corrosion of the component (s) **102**.

(29) In an embodiment, the treating fluid **304** may be configured as a sealant. Accordingly, the treating fluid **304** may preclude contact between a contaminant and component (s) **102**. For example, in an embodiment, corrosion of hot sections of the engine **100**, such as the HP turbine **128**, may stem from sulphates which are stable at relatively high temperatures. In such an embodiment, the sealant may be a high-temp sealant, such as an aluminum oxide-based coating or a phosphate-based coating. By way of further example, in an embodiment, corrosion of cold sections of the engine **100**, such as the fan section **116** and/or the booster compressor **122**, may stem from salts. In such an embodiment, a low-temp sealant, such as a wax or other similar coating may be employed. Additionally, the sealant may be a hydrophobic coating which at least partially repels contaminants, resist corrosion, and/or reduces an aeronautical drag. The hydrophobic coating

may, for example, be a Teflon fluoropolymer coating, a ceramic-based coating (e.g., a silica-based liquid polymer), an aqueous polymer, and/or an aqueous polyurethane.

(30) In an embodiment, the treating fluid **304** may be configured as an inspection aide. As such, the treating flow **304** may, in an embodiment be a dye. In such an embodiment, the treating fluid **304** may be employed to wet the component(s) **102** and then may be rinsed and allowed to dry.

Following the drying, the dye may remain within cracks oxidation sites, fretting sites, and/or other defects thereby facilitating the detection of areas of the component(s) **102** requiring maintenance and/or repair.

(31) It should be appreciated that, in an embodiment, the treating fluid **304** may include compositions configured to affect the component (s) **102** in more than one manner. For example, the treating fluid **304** may include a first composition configured to dissolve a deposit on the component (s) **102** and a second composition configured to seal the component (s) **102**.

(32) In an embodiment, the treating fluid **304** may be configured to affect the component(s) **102** by wetting the component(s) **102**. The wetting of the component(s) **102** may, in an embodiment, include the formation of a liquid film that substantially covers an exposed surface of the component(s) **102**. For example, the system **300** may be configured to form a liquid film over greater than 75% (e.g., at least 80%) of the exposed, inlet-facing surface of the component(s) **102** (e.g., blades and vanes of the HP compressor **124**).

(33) It should be appreciated that the wetting of the component(s) **102** may permit the treating fluid **304**, such as a cleaning fluid, to dissolve/de-bond a portion of the deposit on the component(s) **102**. The dissolution of a portion of the deposit may weaken a bond between the deposited contamination and the surface of the component(s) **102**. In such an embodiment, the weakened bond may permit an additional portion of the deposited contamination to be removed by thermodynamic and/or mechanical forces during a startup of the engine **100**. For example, in an embodiment, a portion of the treating fluid **304** (in atomized droplet form) may penetrate an internal passage of the component(s) **102** to remove a portion of a deposit therein.

(34) In an embodiment, the treating fluid **304** may be configured to be delivered to the engine **100** at a constant rate. In an additional embodiment, the treating fluid **304** may be delivered to the engine **100** at a variable rate. For example, in an embodiment, a first portion of the treating fluid **304** may be delivered at the initiation of the treating cycle so as to rapidly wet the component(s) **102**. The amount of treating fluid **304** delivered to the engine may then be reduced and the wetted component(s) **102** may be permitted to soak. During the soak, a desired level of wetness may be maintained via a second portion of the treating fluid **304**, which may be less than the first portion of the treating fluid **304**. Following at least one soak. A third portion of the treating fluid **304** may be introduced so as to increase the wetness of the component(s) **102**. As is more fully described below, the third portion of the treating fluid **304** may be considered a surge portion.

(35) It should be appreciated that, in an embodiment, the treating fluid **304** may be delivered a single portion without necessitating the introduction of any additional portions of the treating fluid **304**. For example, in an embodiment wherein the treating fluid comprises a sealant, a corrosion treatment, and/or a carbon treatment, a single portion of the treating fluid **304** may be employed to wet the component(s) **102** and achieve the desired treatment thereof. It should further be appreciated that the system **300** may be employed to implement more than one sequential treatment operation. In such an embodiment the treating fluid **304** may be selected to correspond to the desired sequential treatment operation. For example, in an embodiment, the treating fluid **304** may, in a first treatment operation be configured as a cleaning fluid. Following the cleaning of the engine **100**, the treating fluid **304** may be configured for a second treatment operation as a corrosion inhibitor and the engine **100** may be treated in accordance with the systems and methods disclosed herein.

(36) In an embodiment, atomizing a portion of the treating fluid **304** with the delivery assembly **306** may develop the treating mist **312**. The treating mist **312** may include a plurality of atomized

droplets of the treating fluid **304** suspended in a volume of gas, such as a volume of atmospheric air. For example, in an embodiment wherein the treating fluid **304** is a water-based treating fluid, the development of the treating mist **312** may result in a supersaturated vapor component of the treating mist **312** having a water content in excess of that naturally occurring under prevailing ambient conditions. In other words, the atomization of the water of the treating fluid **304** may result in a portion of the atomized droplets evaporating in the air to which the atomized droplets of the treating fluid **304** are introduced, thereby raising the fluid content of the resultant vapor component of the treating mist **312**. The remaining portion of the atomized treating fluid **304** comprising the treating mist **312** may remain as water and/or water-detergent droplets which are entrained in a flow of the vapor component. In an embodiment, the treating mist **312** may have a fluid-to-air mass ratio of at least 0.025 and less than or equal to 5.0 kilograms of treating fluid to kilograms of air. For example, in an embodiment, the fluid-to-air mass ratio may be at least 0.4 and less than 1.0.

(37) It should be appreciated that the effectiveness of the treating mist **312** may be increased by heating the treating fluid **304**, heating the portion of air into which the atomized droplets of the treating fluid **304** are introduced, and/or increasing the delivery pressure of the treating fluid **304** prior to delivery of the treating mist **312** to the annular inlet **120**. Such techniques may increase the fluid content of the treating mist **312** thereby facilitating the wetting of the component(s) **102**. For example, for a water-based treating fluid **304**, the treating fluid **304** may be expelled by the nozzle **310** at a temperature from 95° C. to 260° C. and/or a pressure from 10 kPa to 3,500 kPa. In such an embodiment, the engine **100** to be cleaned may be at a standard atmosphere of 15° C. at 101.325 kPa. Upon delivery to the annular inlet, the treating fluid **304** may encounter lower pressure and temperature conditions than the delivery pressure and temperature, such that the treatment compound vaporizes (if not already a vapor). Because the water-based treatment compound has a delivery pressure (partial pressure) from 103 kPa to 3,447 kPa, and the saturation pressure of a water-based compound at 15° C. is about 1.705 kPa, the resultant vapor may be supersaturated. It should be appreciated that the term “supersaturated,” as used herein, refers to a vapor of a compound that has a higher partial pressure than the vapor pressure of the compound.

(38) In order to increase the temperature of the treating fluid **304** and/or the portion of air into which it is introduced, the system **300** may include at least one heating element **316**. The heating element(s) **316** may be positioned in thermal contact with the treating mist **312** so as to establish an elevated delivery temperature of the treating mist **312** relative to the ambient temperature which increases the vapor content of the treating mist **312**. For example, in an embodiment, the heating element(s) **316** may be positioned in thermal contact with the treating fluid **304** contained within the storage vessel **302**. In an additional embodiment, the heating element(s) **316** may be positioned in thermal contact with an airflow **318** generated by the flow generation assembly **314**. Positioning the heating element(s) **316** in thermal contact with the airflow **318** may increase the temperature of the portion of air into which the atomized droplets of the treating fluid **304** are introduced.

(39) Referring still to FIGS. 1-4, in an embodiment, the atomized droplets of the treating mist **312** may have a median diameter, and therefore mass, which facilitates the atomized droplets remaining suspended in a portion of air within the flow path(s) **104**. At least a portion of the atomized droplets may be entrained in the airflow **318** through the flow path(s) **104** from the annular inlet **120** to the axial position (A) downstream of the HP compressor **124**. It should be appreciated that at various points along the flow path(s) **104**, the airflow **318** may experience a deceleration, such as due to a directional change of the airflow **318** or an effect of drag on a boundary layer of the airflow **318**. As a result of the deceleration, the airflow **318** (or a portion thereof, such as a boundary layer) may lack the necessary energy to retain atomized droplets above a specified mass, as defined by the median diameter. When the energy level of the airflow **318** drops below a given threshold for atomized particles of a given size, the atomized particles may depart the airflow **318** and impact/precipitate within the core engine **114**. Further, for atomized droplets of a given size, the inertial energy of the droplets may result in the atomized droplets departing the airflow **318**.

(40) While the atomized droplets of the treating mist **312** may have a median diameter that facilitates remaining suspended in the portion of air within the flow path(s) **104**, the atomized droplets must also have sufficient mass to violate a flow field of the airflow **318** adjacent to the component(s) **102** and precipitate onto the component(s) **102**. In other words, atomized particles lacking sufficient mass for a given flow may lack sufficient energy to violate a flow field of the airflow **318** (e.g., have a Stokes number of less than 1), may remain entrained in the airflow **318**, and may pass through the gas turbine engine **100** without wetting the component(s) **102**. Therefore, in an embodiment, the atomized particles may have a Stokes number which is greater than 1 (to facilitate the wetting of the booster compressor **122**), while remaining sufficiently low to facilitate wetting the downstream components (e.g., the LP turbine **132**) of the engine **100**.

(41) Contrary to expectations, atomized droplets having a median diameter greater than 10 microns, a low average velocity, but a relatively high local booster inlet velocity may not be suitable for wetting the component(s) **102** at an axial position downstream of the booster compressor **122**. This may be attributed to the atomized droplets having too great of a mass and/or the velocity of the airflow **318** (average and/or local velocity) being too great. For example, approximately 90% or greater of atomized droplets having a median diameter greater than 10 microns for an airflow **318** velocity of at least 0.2 feet per second (fps) and less than or equal to 10 fps may impact a component of the gas turbine engine **100** prior to passing the booster compressor **122**, while the remaining approximately 10% may pass through the gas turbine engine **100** without wetting the component(s) **102** downstream of the booster compressor **122**. Accordingly, in order for at least a portion of the atomized droplets to remain suspended within the flow path(s) **104** from the annular inlet **120** to the axial position (A) while still wetting components **102** adjacent to the annular inlet **120**, the atomized droplets may, in an embodiment, have a median diameter of less than or equal to 10 microns and the airflow **318** may have a velocity at the annular inlet **120** of at least 0.2 fps and less than or equal to 10 fps (e.g., a range of at least 0.2 fps to less than or equal to 2.0 fps). For example, in an embodiment, the airflow **318** may have an average velocity of 1.0 fps.

(42) In an additional embodiment, the atomized droplets may have a median diameter which is at least 2 microns and less than or equal to 5 microns. Unexpectedly, the development of atomized droplets within the range of 2-5 microns inclusive entrained in an airflow having an average inlet annulus velocity of at least 0.2 fps and less than or equal to 2.0 fps, may facilitate both the atomized droplets remaining entrained within the flow path(s) **104** and the wetting of the component(s) **102** downstream of the booster compressor **122**. It may be expected that atomized droplets within this range may be overly controlled by the airflow **318** and therefore pass through the gas turbine engine **100** without wetting the component(s) **102**. However, atomized droplets within the range of 2-5 microns inclusive may, in fact, be impacted by the slowing of the airflow **318** within a boundary layer adjacent to the component(s) **102**. Accordingly, because the airflow within the boundary layer may have a reduced velocity relative to flow fields at a distance from the surface, the atomized droplets adjacent the surface may have sufficient mass to violate the boundary layer flow. As a result, the atomized droplets within the boundary layer may precipitate and wet the surface, while atomized droplets in more distant flow fields remain entrained within the airflow **318**.

(43) In an embodiment, the supportability of the treating mist **312** within the flow path(s) **104** may be enhanced by the formation of a treating mist **312** having atomized droplets with a median diameter greater than or equal to 2 microns and less than or equal to 10 microns (e.g., 2-5 microns). In other words, in an embodiment wherein the water and/or water-detergent droplets have a median diameter in a range between 2 and 10 microns inclusive (particularly between 2 and 5 microns inclusive), the droplets may be entrained in the flow of the vapor component which through the core engine **114**. For example, in an embodiment, a thermal state of the core engine **114** may be less than or equal to 135% of an ambient air absolute temperature in degrees Celsius. In such an embodiment, the median diameter of the atomized droplets may facilitate a portion of the treating

mist **312** remaining suspended within the airflow **318** to an axial position downstream (B) of the installed and assembled gas turbine engine **100**.

(44) In order to develop droplets having the required median diameter, the delivery assembly **306** may atomize a portion of the treating fluid **304**. For example, the delivery assembly **306** may utilize the nozzle **310** to develop the treating mist **312** having atomized droplets of the desired median diameter. The nozzle **310** may utilize at least one orifice and/or the application of ultrasonic energy via an ultrasonic nozzle to atomize the treating fluid **304**. For example, the treating fluid **304** may be drawn through the orifice(s) via a pressure differential across the orifice(s).

Alternatively, the treating fluid **304** may be driven through the orifice(s) by the development of a higher pressure within the storage vessel **302** than at the orifice(s). The higher pressure may be developed via a pump, a compressed air source **326**, and/or heating. In at least one embodiment, the compressed air source **326** may also be configured as the flow generation assembly **314** to facilitate the passage of the treating mist **312** along the flow path(s) **104**.

(45) In an additional embodiment, the nozzle **310** may be configured as an ultrasonic transducer. In such an embodiment, the nozzle **310** may interact with a portion of the treating fluid **304** so as to atomize a portion of the treating fluid **304**. The resultant atomized portion may be drawn from the treating fluid **304** by the flow generation assembly **314** for delivery to the annular inlet **120**.

(46) In an embodiment, the nozzle **310** may be configured as a vibrating mesh nebulizer. The vibrating mesh nebulizer may be configured to develop droplets having a median diameter of less than or equal to 10 microns. For example, in an embodiment, the vibrating mesh nebulizer may be employed to develop the treating mist **312** with atomized droplets having a median diameter of at least 2 microns and less than or equal to 5 microns so as to facilitate both the suspension of a portion of the atomized droplets within the flow path(s) **104** to at least the axial position downstream (B) and the wetting of at least 80% of the exposed, inlet-facing surface of the component(s) **102**.

(47) In an embodiment, the nozzle **310** may utilize surface acoustic wave nebulization to develop droplets having a median diameter of less than or equal to 10 microns. For example, in an embodiment, the vibrating mesh nebulizer may be employed to develop the treating mist **312** with atomized droplets having a median diameter of at least 2 microns and less than or equal to 5 microns so as to facilitate both the suspension of a portion of the atomized droplets within the flow path(s) **104** to at least the axial position downstream (B) and the wetting of at least 80% of the exposed, inlet-facing surface of the component(s) **102**.

(48) As depicted in FIG. 2, the nozzle **310** may be positioned upstream of the fan section **116** and operably coupled to the annular inlet **120** of the core engine **114**. In an additional embodiment, the nozzle **310** may be one of an array of nozzles **320**. For example, the array of nozzles **320** may include four or more nozzles **310**. As depicted in FIG. 1, the array of nozzles **320** may be arranged to interface with the fan section **116** so as to position an outlet **322** of each nozzle **310** at an axial location between the fan section **116** and the annular inlet **120**. In such an embodiment, the number of nozzles **310** of the array of nozzles **320** may correspond to the number of spaces between the fan blades **144** such that at least one nozzle **310** is inserted between each pair of fan blades **144**. In an embodiment, the nozzles **310** may be operably coupled to the annular inlet **120** without necessitating contact with the engine **100**.

(49) It should be appreciated that, in an embodiment, the nozzle **310** may be positioned at the axial position downstream (B) of the installed and assembled gas turbine engine **100**. In such an embodiment, the airflow **318** may be configured to progress from the point-of-departure **136** toward annular inlet **120** (e.g., the airflow **318** may be reversed relative to the nominal passage of air through the gas turbine engine **100** during operation). In such an embodiment, a portion of the treating mist **312** with atomized droplets having a median diameter of at least 2 microns and less than or equal to 5 microns may be suspended within the flow path(s) **104** to at least an axial position axially upstream of the HP compressor **124**. It should be appreciated that the introduction

of the treating mist **312** at the point-of-departure **136** may facilitate the treatment of targeted components of the gas turbine engine **100**.

(50) In an embodiment wherein the delivery assembly **306** includes the array of nozzles **320**, modifying the number of nozzles **310** employed to atomize the portion of the treating fluid **304** may affect the concentration of atomized droplets within a specified portion **330** of the flow path(s) **104**. As such, modifying the number of nozzles **310** may establish a treating mist volume. For example, in an embodiment wherein the entire array of nozzles **320** is activated, the concentration of atomized droplets may be greater than in an embodiment wherein the majority of the array of nozzles **320** are idle.

(51) Referring now particularly to FIG. **3**, in an embodiment, the delivery assembly **306** may include a duct portion **324** communicatively coupled between the core engine **114** and the storage vessel **302**. In such an embodiment, the atomization of the treating fluid **304** may occur in, or adjacent to, the storage vessel **302**. The atomized droplets may enter the duct portion **324** of the delivery assembly **306** and be delivered to the annular inlet **120** by the flow generating assembly **314**. In an embodiment, the flow generating assembly **314** may be incorporated into the duct portion **324**. Additionally, in an embodiment, duct portion **324** may incorporate the heating element(s) **316**.

(52) Referring again to FIGS. **1-3**, in an embodiment, the system **300** may include the flow generation assembly **314**. The flow generation assembly **314** may facilitate the passage of the treating mist **312** along the flow of path **104** by developing the airflow **318**. In an embodiment, the flow generation assembly **314** may be configured to drive or draw a portion of atmospheric air so as to generate the airflow **318**. In such an embodiment, the flow generation assembly **314** may, for example, be configured as a fan. Accordingly, the flow generation assembly may be positioned upstream of the annular inlet **120** when configured to accelerate or drive the airflow **318** and may be positioned downstream of the engine **100** when configured to draw the treating mist **312** through the flow path(s) **104**.

(53) In an embodiment, the flow generation assembly **314** may be configured to establish a pressure differential between the annular inlet **120** and the point-of-departure **136**. For example, in an embodiment, the flow generation assembly **314** may be configured as the compressed air source **326**. In such an embodiment, the compressed air may not only drive the treating fluid **304** through the nozzle **310**, but the venting of the compressed air through the nozzle **310** may create a region of increased pressure in fluid communication with the annular inlet **120**. The region of increased pressure may drive the treating mist **312** through the flow path(s) **104**. In a further embodiment, a high-pressure region may be established adjacent to the annular inlet **120** via the heating of a portion of atmospheric air adjacent to the annular inlet **120**. In yet a further embodiment, the flow generation assembly **314** may be positioned downstream of the engine **100** and configured to establish a low-pressure region adjacent to the point-of-departure **136**.

(54) In an embodiment, the system **300** may include an extraction element **346**. The extraction element **346** may be positioned at the point-of-departure **136** (e.g. adjacent the axial position downstream (B)). The extraction element **346** may be configured to extract a residual portion of the treating mist **312** from the airflow **318**. For example, the extraction element **346** may include a condenser, a mesh, and/or equivalent structure. In an embodiment, the extraction element **346** may be an independent element of the system **300**. However, in an embodiment, the extraction element **346** may be incorporated as a component of the flow generation assembly **314** when the flow generation assembly **314** is positioned at the point-of-departure **136**.

(55) The extraction element **346** may, in an embodiment, store the extracted residual portion of the cleaning mist **312**. In such an embodiment, the extracted residual portion of the cleaning mist **312** may be subsequently disposed of, reused, or treated and reused. In an additional embodiment, the extraction element **346** may be fluidly coupled to the storage vessel **302** thereby facilitating the recirculation of at least a portion of the treating fluid **304**. It should be appreciated that employing

the extraction element **346** may increase the safety and/or efficiency of the system **300**. For example, extracting the residual portion of the treating mist **312** at the point-of-departure **136** may protect personnel from inhalation hazards, reduce environmental contamination, and/or reduce contamination of airframe parts by the treating fluid **302**.

(56) Referring still to FIGS. **1-3**, in an embodiment, the flow generation assembly **314** of the system **300** may be configured as a heat source positioned downstream of the point-of-departure **136**. In such an embodiment, the flow generation assembly **314** may heat the portion of atmospheric air adjacent to the point-of-departure **136**. This heating of the atmospheric air may establish the airflow **318** as a convection current through the core engine **114**. The treating mist **312** may be drawn through the flow path(s) **104** via the convection current.

(57) Referring again to FIG. **1**, in an embodiment, the system **300** may include a control unit **308**. The control unit **308** may, in an embodiment, include a plurality of controls configured to permit an operator to employ the system **300** to treat the component(s) **102** of the installed and assembled gas turbine engine **100** (e.g., the assembled, on-wing gas turbine engine). For example, in an embodiment, the controls may be configured to permit an operator to employ the system **300** to remove a deposit from the component(s) **102** of the installed and assembled gas turbine engine **100**. In at least one embodiment, the control unit may include various readouts configured to provide the operator with information concerning the treating of the engine **100** and various manual controls configured to provide the operator with the necessary degree of control over the methods described herein to remove deposits from the component(s) **102**. In an additional embodiment, the control unit **308** may also include a controller **400**. The controller **400** may also be configured to implement the methods discussed herein to treat the component(s) **102** of the engine **100**. It should be appreciated that the controller **400** may be employed in combination with various manual controls and displays to facilitate an operator's control of the methods described herein.

(58) In an embodiment, treating (e.g. cleaning) the component(s) **102** may include the operator/controller **400** obtaining environmental data indicative of the environmental conditions affecting the installed and assembled gas turbine engine. For example, the environmental data may include an ambient temperature, an ambient pressure, and an ambient humidity affecting the engine **100**. In an embodiment, the environmental data may also include data indicative of the type of suspended atmospheric particulate (e.g. dust-reacted and unreacted, sand, etc.), an atmospheric particulate concentration, and/or an atmospheric particulate size for an operating environment of the engine **100**. In other words, the environmental data may include data on the type and severity of contaminants encountered during the engine's operations. It should be appreciated that the information concerning the encountered particulates may be utilized by the operator/controller **400** to determine a treating fluid composition, treating operation duration and/or treating operation frequency. For example, the information concerning the encountered particulates may be utilized by the operator/controller **400** to determine a cleaning fluid composition which may effectively remove deposits from or prevent the corrosion of the component (s) **102**. It should further be appreciated that the environmental data may be obtained from any suitable source, such as a plurality of sensors, an external provider, and/or a lookup table.

(59) The operator/controller **400** may also obtain data indicative of the thermal state of the core engine **114**. The thermal state may indicate a difference between the temperature of components within the core engine **114** and the ambient temperature. The data indicative of the thermal state may be obtained via at least one sensor and/or via a lookup table. For example, the thermal state may be determined relative to an elapsed time since engine shutdown under ambient atmospheric conditions. Based on the environmental data and the data indicative of the thermal state, the operator/controller **400** may establish the treating mist volume delivered to the annular inlet **120**. For example, establishing the treating mist volume may include actuating a nozzle **310** to establish a concentration of atomized droplets within a specified portion **330** of the flow path(s) **104** at a specified time interval. Additionally, establishing the treating mist volume may also include

establishing a treating mist flow rate within the flow path(s) **104**. Establishing the treating mist flow rate may include establishing/modifying the velocity of the airflow **318**. It should be appreciated that the velocity of the airflow **318** in conjunction with the number of nozzles **310** actuated may affect the resultant density of the treating mist **312** within the flow path(s) **104**.

(60) Referring still to FIG. **1**, in an embodiment, the system **300** may also include an electronic sensor **328** (e.g., a humidity sensor, a lidar unit, an anemometer, and/or any other suitable sensor for detecting water/water vapor). The electronic sensor **328** may be communicatively coupled to the control unit **308** and may be positioned at the point-of-departure **136** from the assembled, on-wing, gas turbine engine **100**. The electronic sensor **328** may monitor an absolute humidity level and/or quantity of liquid water droplets at the point-of-departure **136**. The absolute humidity level/water droplet quantity may be utilized by the operator/controller **400** to determine a percentage of the delivered treating mist **312** remaining suspended at the point-of-departure **136** from the engine **100**. For example, the absolute humidity level at the point-of-departure **136** may, when corrected for the ambient temperature, be indicative of the fluid content of the airflow **318** at the point-of-departure **136**. The fluid content may, in turn, be indicative of a percentage of treating mist **312** introduced to the annular inlet **120** which remains suspended at the point-of-departure **136**.

(61) Utilizing the determined percentage of the treating mist **312** remaining suspended at the point-of-departure **136**, the operator/controller **400** may, in an embodiment, adjust the treating mist volume delivered to the annular inlet **120** so as to achieve a desired level of wetting of the component(s) **102**. For example, in an embodiment wherein the thermal state of the engine **100** is relatively high when the system **300** is activated, a significant percentage of the treating mist **312** may be converted to vapor in cooling the core engine **114**. In such an embodiment, the absolute humidity level at the point-of-departure **136** may be relatively close to the ambient humidity level and may indicate a need to increase the treating mist volume to achieve the desired level of wetting of the component(s) **102**. In a further embodiment, the absolute humidity level at the point-of-departure **136** may be significantly higher than the ambient humidity level, thus indicating an excessive amount of treating mist **312** remains suspended at the point-of-departure **136**. In such an embodiment, the operator/controller **400** may reduce the treating mist volume and/or the velocity of airflow **318** so as to improve the efficiency of the system **300**. Accordingly, in an embodiment, a treating cycle of the engine **100** may consume less than 57 liters of treating fluid **304**. For example, the system **300** may, in an embodiment, be employed to clean the engine **100** while consuming less than 57 liters of treating fluid **304** configured as a cleaning fluid.

(62) In an embodiment, such as particularly depicted in FIGS. **2** and **3**, the system **300** may be configured to determine a first volume of treating fluid **304** atomized and delivered as the treating mist **312** to the annular inlet. The first volume may, for example, be determined via the monitoring of a dispersal rate of the treating fluid **304** from the storage vessel **302**. In an embodiment, the system **300** may also be configured to determine a second volume **332** of treating fluid **304** suspended at the point-of-departure **136** based on the retained total water level as may be indicated by the monitored absolute humidity level. Computing the difference between the first and second volumes may indicate the portion of the first volume of the treating fluid **304** precipitated/impacted onto the component(s) **102**.

(63) In an embodiment, the operator/controller **400** may adjust the treating mist volume delivered to the annular inlet **120** so that the second volume is less than or equal to 35% of the first volume. In other words, the system **300** may, in an embodiment, be configured to deliver the treating mist **312** at a treating mist volume and flowrate at which 65% or more of the treating mist **312** is deposited within the core engine **114**. This may, for example, be desirable in an embodiment wherein the engine **100** is in a cold, thermally stable condition. However, for a hot engine **100**, excess treating fluid **30** may be introduced to the engine **100** so as to accelerate the cooling and wetting of the surfaces of the component(s) **102**. Accordingly, the second volume may exceed 35%

of the first volume. It should be appreciated that ensuring that no more than 35% of the treating mist **312** passes completely through the core engine **114** once the core engine **114** is cooled may ensure that the system **300** is operating in an efficient manner. For example, the efficient manner may be defined by the execution of a treating cycle within 15 minutes which consumes less than 57 liters of treating fluid **304**.

(64) Referring again to FIG. **1**, in an embodiment, the system **300** may include at least one blocking element **334** circumscribing the annular inlet **120**. Additionally, in an embodiment, the blocking element(s) **334** may circumscribe the array of nozzles **320**. The blocking element(s) **334** may at least partially occlude the alternative flow path(s) **148**. It should be appreciated that occluding the alternative flow path(s) **148**, may increase the efficiency of the system **300** by reducing or eliminating the portion of the treating mist **312** which may bypass the flow path(s) **104**.

(65) Referring now to FIG. **4**, in an embodiment, the system **300** may be configured for employment on the engine **100** while the aircraft **336** is parked at a gate **338** of an airport **340**. Accordingly, the engine **100** may be treated utilizing the system **300** following a landing **342** or prior to a takeoff **344** without requiring that the aircraft **336** be moved to a maintenance area of the airport **340**. For example, the engine **100** may be cleaned utilizing the system **300** following a landing **342** or prior to a takeoff **344** without requiring that the aircraft **336** be moved to a maintenance area of the airport **340**. It should be appreciated that performing the treating (e.g. cleaning) operations described herein while the aircraft **336** is located at the gate **338** may reduce or eliminate the amount of time the aircraft **336** is removed from service for engine treating operations.

(66) In an embodiment, the system **300** may be configured to execute a treating cycle which has a duration of 30 minutes or less and a consumption of less than 120 liters. For example, the system **300** may be employed to dissolve a portion of the deposits on the component(s) **102** within 15 minutes of the cycle initiation with a consumption of less than 57 liters.

(67) In an embodiment, following the conclusion of the treating cycle, the delivery assembly **306** may be operably decoupled from the annular inlet **120**. However, as the treating cycle disclosed herein does not require a rinse cycle, the component(s) **102** may remain in a wetted condition following the operable decoupling of the delivery assembly **306**.

(68) In an embodiment, the absence of a requirement to rinse the engine may be leveraged by supplying a surge portion of the treating mist **312** to the core engine **114** prior to operably decoupling the delivery assembly **306**. In such an embodiment, a soak period may be established following the operable decoupling of the delivery assembly **306** during which the treating fluid **304** continues affecting the component(s) **102**. For example, the soak period may facilitate the continued affecting of the deposits on the component(s) **102** by the treating fluid **304** following the operable decoupling of the delivery assembly **306**. Accordingly, the surge portion may increase the wetness of the component(s) **102**, thereby increasing the effectiveness of the soak period.

(69) In an embodiment, following the introduction of the treatment fluid **304**, the engine **100** may be placed into a low-power, low-temperature operating condition for a specified duration in order to complete the treatment operation. For example, placing the engine **100** into a low-power, low temp operating condition for 15-60 minutes may melt, cure, and/or oxidize the deposited treatment fluid **304**. The melting, curing, and/or oxidizing may ensure that the deposited treatment fluid **304** is stable on the component(s) **102** prior to the engine **100** being brought to full power. It should be appreciated that bringing the engine **100** to full power prior to the melting, curing, and/or oxidizing may aerodynamically and/or through rapid thermal shock remove the treatment fluid **304** before it is bonded and/or the treatment operation is complete.

(70) The ability to execute a treating cycle within 30 minutes or less while the aircraft **336** is parked at the gate **338** may facilitate an increased treating frequency relative to current engine treating approaches. As such, in an embodiment, the treating of the component(s) **102** may be repeated at least once every 30 days and/or engine cycles. For example, the ability to execute a

cleaning cycle within 30 minutes or less while the aircraft **336** is parked at the gate **338** may facilitate an increased cleaning frequency relative to current engine cleaning approaches. As such, in an embodiment, the removing of the deposit from the component(s) **102** may be repeated at least once every 30 days and/or engine cycles. Accordingly, in an embodiment, the system **300** may be employed at least once every 7 days to treat/remove deposits from the component(s) **102**. It should be appreciated that an engine cycle may correspond to a 24-hour period, a takeoff/landing sequence, and/or a specified quantity of engine operating hours.

(71) In an embodiment, the data on the type and severity of contaminants encountered during the engine's operations may be utilized to modify the treating interval. For example, the type and/or severity of the contaminants encountered during the engine's operations may be high and may indicate that a reduced treating (e.g., cleaning) interval may be warranted. Such particulate conditions may, for example, be encountered when operating the engine **100** in dusty/desert environments and may justify removing the deposits from the component(s) **102** with the system **300** on a nightly basis.

(72) It should be appreciated that frequent treatments may preserve the efficiency and lifespan of the engine **100**. It should also be appreciated that the more frequent treatments, such as cleanings, may, individually, remove a smaller percentage of the deposits on the component(s) **102** relative to traditional water/foam treatment/washes. However, the effects of frequent treatments may accumulate such that the combined effect of the frequent treatments is cumulatively greater than any single water/foam treatment. As a result, the shortened intervals between treatments/clinics may result in the engine being operated in a less-fouled condition for a greater percentage of the time than is achievable under traditional approaches.

(73) Referring now to FIG. **6**, a flow diagram of one embodiment of a method **504** treating at least one component of an installed and assembled gas turbine engine is illustrated. The method **500** may be implemented using, for instance, the system **300** discussed above with references to FIGS. **1-4**. FIG. **6** depicts steps performed in a particular order for purposes of illustration and discussion. Those of ordinary skill in the art, using the disclosures provided herein, will understand that various steps of the method **500**, or any of the methods disclosed herein, may be adapted, modified, rearranged, performed simultaneously, or modified in various ways without deviating from the scope of the present disclosure.

(74) As shown at (**502**), the method **500** may include operably coupling a delivery assembly to an annular inlet of the core gas turbine engine. The delivery assembly being coupled to a control unit and a storage vessel containing a treating fluid. As shown at (**504**), the method **500** may include atomizing a portion of the treating fluid with the delivery assembly to develop a treating mist. The treating mist may include a plurality of atomized droplets. Additionally, as shown at (**506**), the method **500** may include suspending the atomized droplets of the treating mist within an airflow within at least one flow path of the core gas turbine engine. The treating mist may occupy a cross-sectional area of the at least one flow path to establish a simultaneous cross-sectional contact therein. At least a portion of the atomized droplets may remain suspended within the at least one flow path from the annular inlet to an axial position downstream of a compressor of the gas turbine engine. As shown at (**508**), the method **500** may include impacting or precipitating a portion of the treating mist onto the at least one component so as to wet at least 80% of an exposed surface of the at least one component.

(75) FIG. **5** provides a block diagram of an exemplary controller **400** that may be used to implement the methods and systems described herein according to exemplary embodiments of the present disclosure. Though described below as a computing system, it should be appreciated that in some embodiments, the controller may be an analog system or an electrical system that does not include a computing device. As shown, the computing system **400** may include one or more computing device(s) **402**. The one or more computing device(s) **402** may include one or more processor(s) **404** and one or more memory device(s) **406**. The one or more processor(s) **404** may

include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, or other suitable processing device. The one or more memory device(s) **406** may include one or more computer-readable media, including, but not limited to, non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, or other memory devices.

(76) The one or more memory device(s) **406** may store information accessible by the one or more processor(s) **404**, including computer-readable instructions **408** that may be executed by the one or more processor(s) **404**. The instructions **408** may be any set of instructions that when executed by the one or more processor(s) **404**, cause the one or more processor(s) **404** to perform operations. The instructions **408** may be software written in any suitable programming language or may be implemented in hardware. In some embodiments, the instructions **408** may be executed by the one or more processor(s) **404** to cause the one or more processor(s) **404** to perform operations, such as implementing one or more of the processes mentioned above.

(77) The memory device(s) **404** may further store data **410** that may be accessed by the processor(s) **404**. The data **410** may include one or more table(s), function(s), algorithm(s), model(s), equation(s), etc. according to example embodiments of the present disclosure.

(78) The one or more computing device(s) **402** may also include a communication interface **412** used to communicate, for example, with the other component(s) of system. The communication interface **412** may include any suitable component(s) for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, or other suitable component(s).

(79) The technology discussed herein makes reference to computer-based systems and actions taken by and information sent to and from computer-based systems. One of ordinary skill in the art will recognize that the inherent flexibility of computer-based systems allows for a great variety of possible configurations, combinations, and divisions of tasks and functionality between and among components. For instance, processes discussed herein may be implemented using a single computing device or multiple computing devices working in combination. Databases, memory, instructions, and applications may be implemented on a single system or distributed across multiple systems. Distributed components may operate sequentially or in parallel.

(80) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

(81) Further aspects of the invention are provided by the subject matter of the following clauses:
Clause 1. A method for treating at least one component of an installed and assembled gas turbine engine, the gas turbine engine comprising a core gas turbine engine, the core gas turbine engine having an absence of powered rotation during the treating, the method comprising: operably coupling a delivery assembly to an annular inlet of the core gas turbine engine, the delivery assembly being coupled to a control unit and a storage vessel containing a treating fluid; atomizing a portion of the treating fluid with the delivery assembly to develop a treating mist, the treating mist comprising a plurality of atomized droplets; suspending the atomized droplets of the treating mist within an airflow within at least one flow path of the core gas turbine engine, wherein the treating mist occupies a cross-sectional area of the at least one flow path to establish a simultaneous cross-sectional contact therein, wherein at least a portion of the atomized droplets remain suspended within the at least one flow path from the annular inlet to an axial position downstream of a high-pressure compressor of the core gas turbine engine; and impacting or precipitating a portion of the treating mist onto the at least one component so as to wet at least 80% of an exposed,

inlet-facing surface of the at least one component. Clause 2. The method of clause 1, wherein the atomized droplets have a median diameter of less than or equal to 5 microns and the treating mist has a fluid-to-air mass ratio of at least 0.025 and less than or equal to 5.0 kilograms of treating fluid to kilograms of air. Clause 3. The method of any preceding clause, wherein a thermal state of the core gas turbine engine is less than or equal to 135% of an ambient air temperature, and wherein the axial position downstream of the compressor is an axial position downstream of the installed and assembled gas turbine engine. Clause 4. The method of any preceding clause, further comprising: establishing an elevated delivery temperature of the treating mist which increases a vapor content of the treating mist within the at least one flow path. Clause 5. The method of any preceding clause, further comprising: supplying a surge portion of the treating mist to the core gas turbine engine; operably decoupling the delivery assembly from the annular inlet; and establishing a soak period during which the treating fluid affects at least one of the at least one component and a deposit thereon. Clause 6. The method of any preceding clause, wherein the delivery assembly comprises an array of nozzles, wherein each nozzle of the array of nozzles is configured to develop atomized droplets having a median diameter of less than or equal to 5 microns, the method further comprising: actuating at least one nozzle of the array of nozzles to establish a treating mist volume, wherein the treating mist volume comprises a concentration of atomized droplets within a specified portion of the at least one flow path, the treating mist volume being characterized by a fluid-to-air mass ratio of at least 0.025 and less than or equal to 5.0 kilograms of treating fluid to kilograms of air. Clause 7. The method of any preceding clause, further comprising: obtaining environmental data, the environmental data comprising an ambient temperature, an ambient pressure, and an ambient humidity affecting the installed and assembled gas turbine engine; obtaining data indicative of a thermal state of the core gas turbine engine; and based on the environmental data and the data indicative of the thermal state of the core gas turbine engine, establishing the treating mist volume delivered to the annular inlet, wherein establishing the treating mist volume further comprises establishing a treating mist flow rate within the at least one flow path. Clause 8. The method of any preceding clause, wherein the environmental data further comprises data indicative of at least one of a type of suspended atmospheric particulate, an atmospheric particulate concentration, and an atmospheric particulate size for an operating environment of the installed and assembled gas turbine engine. Clause 9. The method of any preceding clause, wherein further comprising: monitoring an absolute humidity level at a point-of-departure from the installed and assembled gas turbine engine; utilizing the monitored absolute humidity level to determine a percentage of the delivered treating mist remaining suspended at the point-of-departure from the installed and assembled gas turbine engine; and adjusting the treating mist volume delivered to the annular inlet based on the determined percentage so as to achieve a desired level of wetting of the at least one component. Clause 10. The method of any preceding clause, further comprising: determining a first volume of treating fluid atomized and delivered as the treating mist to the annular inlet; determining a second volume of treating fluid suspended at the point-of-departure from the installed and assembled gas turbine engine based on the monitored humidity level; computing the portion of the first volume of treating fluid precipitated onto the at least one component based on a difference between the first and second volumes; and adjusting the treating mist volume delivered to the annular inlet so that the second volume is less than or equal to 35% of the first volume. Clause 11. The method of any preceding clause, wherein the delivery assembly comprises an array of nozzles, and wherein operably coupling the delivery assembly to the annular inlet further comprises: positioning the array of nozzles within the fan section so that an outlet of each nozzle of the array of nozzles is arranged at an axial location between the fan section and the annular inlet of the core gas turbine engine; and circumscribing the array of nozzles with at least one blocking element, wherein the at least one blocking element at least partially occludes an alternative flow path. Clause 12. The method of any preceding clause, further comprising: establishing a negative pressure differential between the annular inlet and a point-of-departure from

the installed and assembled gas turbine engine, wherein the negative pressure differential draws a portion of the treating mist through the core gas turbine engine. Clause 13. The method of any preceding clause, wherein operably coupling the delivery assembly to the annular inlet of the core gas turbine engine further comprises operably coupling the delivery assembly to the annular inlet of the core gas turbine engine of an aircraft parked at a gate of an airport. Clause 14. The method of any preceding clause, wherein treating the at least one component of the installed and assembled gas turbine engine is repeated at least once every 30 days. Clause 15. The method of any preceding clause, further comprising: establishing a treating cycle, wherein the treating cycle has a duration of 30 minutes or less; and operably decoupling the delivery assembly from the annular inlet by a conclusion of the treating cycle. Clause 16. The method of any preceding clause, wherein atomizing a portion of the treating fluid comprises atomizing less than 120 liters of treating fluid. Clause 17. The method of any preceding clause, wherein the treating fluid comprises a cleaning fluid, wherein the treating mist is a cleaning mist, and wherein the method further comprises: dissolving at least a portion of a deposit on the at least one component so as to remove the deposit. Clause 18. The method of any preceding clause, wherein the treating fluid comprises a corrosion inhibitor, the corrosion inhibitor being a catalyst configured to react with at least one identified contaminant so as to inhibit corrosion of the at least one component. Clause 19. The method of any preceding clause, wherein the treating fluid comprises a sealant configured to preclude a contact between a contaminant and the at least one component. Clause 20. The method of any preceding clause, wherein atomizing the portion of the treating fluid further comprises: atomizing the portion of the treating fluid via a vibrating mesh nebulizer so as to develop the plurality of atomized droplets having a median diameter of at least 2 microns and less than or equal to 5 microns. Clause 21. The method of any preceding clause, further comprising: treating an internal passage of the at least one component via the penetration of the internal passage by the atomized droplets, wherein penetration of the internal passage removes at least a portion of a deposit therein.

Claims

1. A system for treating at least one component of an installed and assembled gas turbine engine, the gas turbine engine comprising a core gas turbine engine and a fan section, the system comprising: a storage vessel configured for containing a treating fluid; a control unit; a delivery assembly operably coupled to the storage vessel and the control unit, the delivery assembly being configured to atomize a portion of the treating fluid to develop a treating mist comprising a plurality of atomized droplets, the atomized droplets having a median diameter facilitating suspension of the atomized droplets within at least one flow path of the core gas turbine engine, wherein the median diameter of the atomized droplets is less than or equal to 10 microns, the treating mist configured to occupy a cross-sectional area of the at least one flow path of the core gas turbine engine to establish a simultaneous cross-sectional contact therein; a flow generation assembly, the flow generation assembly configured to facilitate a passage of the treating mist along the at least one flow path of the core gas turbine engine; and a humidity sensor configured to be positioned at a point-of-departure from the gas turbine engine and communicatively coupled to the control unit, wherein the humidity sensor is configured to monitor a humidity level at the point-of-departure from the gas turbine engine, wherein the control unit is configured to utilize the monitored humidity level to determine a percentage of the treating mist remaining suspended at the point-of-departure from the gas turbine engine, and wherein the control unit is configured to adjust a treating mist volume delivered to an annular inlet of the core gas turbine engine based on the determined percentage so as to achieve a desired level of wetting of the at least one component.
2. The system of claim 1, further comprising: a heating element positioned to come into thermal contact with the treating fluid so as to establish an elevated delivery temperature of the treating mist which increases a vapor content of the treating mist.

3. The system of claim 1, further comprising: a heating element positioned to come into thermal contact with an airflow generated by the flow generation assembly.
4. The system of claim 1, wherein the treating mist volume is characterized by a fluid-to-air mass ratio of at least 0.05 and less than or equal to 5.0 kilograms of the treating fluid to kilograms of air.
5. The system of claim 1, wherein the atomized droplets have a median diameter of less than or equal to 5 microns and a treating mist volume is characterized by a fluid-to-air mass ratio of at least 0.025 and less than or equal to 5.0 kilograms of the treating fluid to kilograms of air.
6. The system of claim 1, wherein the delivery assembly comprises an array of nozzles, wherein the array of nozzles is arranged to interface with the fan section so as to position an outlet of each nozzle of the array of nozzles at an axial location between the fan section and an annular inlet of the core gas turbine engine, the system further comprising: at least one blocking element circumscribing the array of nozzles, wherein the at least one blocking element at least partially occludes an alternative flow path.
7. The system of claim 1, wherein the delivery assembly comprises an array of nozzles, wherein individual nozzles of the array of nozzles are configured to develop atomized droplets having a median diameter of less than or equal to 5 microns, and at least one nozzle of the array of nozzles is configured to establish a treating mist volume, wherein the treating mist volume comprises a concentration of atomized droplets within a specified portion of the at least one flow path, the treating mist volume being characterized by a fluid-to-air mass ratio of at least 0.025 and less than or equal to 5.0 kilograms of treating fluid to kilograms of air.
8. The system of claim 7, wherein the control unit is configured to: obtain environmental data, the environmental data comprising an ambient temperature, an ambient pressure, and an ambient humidity affecting the installed and assembled gas turbine engine; obtain data indicative of a thermal state of the core gas turbine engine; and based on the environmental data and the data indicative of the thermal state of the core gas turbine engine, establish the treating mist volume delivered and cause the array of nozzles to establish a treating mist flow rate within the at least one flow path.
9. The system of claim 8, wherein the control unit is configured to: monitor an absolute humidity level at a point-of-departure from the installed and assembled gas turbine engine; utilize the monitored absolute humidity level to determine a percentage of the delivered treating mist remaining suspended at the point-of-departure from the installed and assembled gas turbine engine; and adjust the treating mist volume delivered to an annular inlet of the core gas turbine engine based on the determined percentage so as to achieve a desired level of wetting of the at least one component.
10. The system of claim 9, wherein the control unit is further configured to: determine a first volume of treating fluid atomized and delivered as the treating mist to the annular inlet; determine a second volume of treating fluid suspended at the point-of-departure from the installed and assembled gas turbine engine based on the monitored humidity level; compute the portion of the first volume of treating fluid precipitated onto the at least one component based on a difference between the first and second volumes; and adjust the treating mist volume delivered to the annular inlet so that the second volume is less than or equal to 35% of the first volume.
11. The system of claim 1, wherein the flow generation assembly is configured to establish a negative pressure differential between an annular inlet of the core gas turbine engine and a point-of-departure from the installed and assembled gas turbine engine, wherein the negative pressure differential is configured to draw a portion of the treating mist through the core gas turbine engine.
12. The system of claim 1, wherein the treating fluid comprises a corrosion inhibitor, the corrosion inhibitor being a catalyst configured to react with at least one identified contaminant so as to inhibit corrosion of the at least one component.
13. The system of claim 1, wherein the treating fluid comprises a sealant configured to preclude a contact between a contaminant and the at least one component.

14. The system of claim 1, wherein the delivery assembly comprises a mesh nebulizer that is configured to vibrate so as to develop the plurality of atomized droplets having a median diameter of at least 2 microns and less than or equal to 5 microns.
 15. The system of claim 1, wherein the delivery assembly comprises at least one ultrasonic transducer.
 16. The system of claim 1, wherein the flow generation assembly comprises at least one of a pump and a compressed air source.
 17. The system of claim 1, wherein the delivery assembly comprises a duct portion configured to be communicatively coupled to the storage vessel and the flow generation assembly is incorporated within the duct portion.
 18. The system of claim 17, wherein at least one nozzle is disposed within the duct portion.
 19. The system of claim 1, further comprising an extraction element configured to be positioned at a point-of-departure from the gas turbine engine for extracting a residual portion of the treating mist.
 20. The system of claim 19, wherein the extraction element includes at least a component of the flow generation assembly.
 21. The system of claim 1, wherein the delivery assembly is configured to develop the treating mist within the at least one flow path of the core gas turbine engine from a point-of-departure from the gas turbine engine toward an annular inlet of the core gas turbine engine.
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